

Project data request document

Client: STADIOcare Group

Table of contents

Introduction	2
Data request 1: History of patient admissions and discharge.....	2
Data request 2: Patient demographics	3
Data request 3: Medical history and clinical data.....	3
Data request 4: Medication and treatment history	3
Data request 5: Follow-up and outpatient appointments.....	4
Data request 6: Claim and billing information.....	4
Data request 7: Procedure information	5
Data request 8: Medical scheme rules (if available)	6

Introduction

According to brief from STADIOcare the readmission rate has increased from 8.9% to 11.3%, by reducing readmissions directly supports the company's strategic objective of keeping patients out of hospital when appropriate while improving patient outcomes and reducing costs.

These tables below will allow me to identify patterns associated with readmissions or claim rejections to allow STADIOcare to intervene proactively, reduce avoidable readmissions and reduced short paid claims or claim rejections.

Data request 1: History of patient admissions and discharge

Column Name	Data Type	Example	Additional Information
Patient_ID	String	PAT123456	Unique patient identifier
Admission_ID	String	ADM987654	Unique admission record
Admission_Date	Datetime	2025-05-03 08:15:00	
Discharge_Date	Datetime	2025-05-07 14:30:00	
Length_of_Stay	Numeric, Integer	4	Number of days admitted
Hospital_ID	String	JHB001	Hospital where patient was treated
Ward_Type	String, Categorical	ICU	ICU, High Care, General, Surgical, etc.
Admission_Type	String, Categorical	Emergency	Emergency, Elective, Transfer
Procedure_Code	String	KNEE001	Standard procedure code
Procedure_Description	String	Knee Replacement	Full procedure name
Discharge_Status	String	Home	Home, Transfer, Deceased, Other
Previous_Admissions	Numeric, Integer	3	Number of prior admissions
Readmitted_30_Days	Boolean	Yes	May be used as Target variable for model

Data request 2: Patient demographics

Column Name	Data Type	Example	Additional Information
Patient_ID	String	PAT123456	This must match admissions table
Age	Numeric, Integer	68	Patient age in years
Gender	String, Categorical	Female	Male, Female, Other
Province	String	GP	
Medical_Scheme	String	Discovery	Medical aid/provider name
Membership_Type	String	Comprehensive	Plan type
Employment_Status	String	Retired	

Data request 3: Medical history and clinical data

Column Name	Data Type	Example	Additional Information
Patient_ID	String	PAT123456	
Chronic_Diabetes	Boolean	Yes	Yes/No
Chronic_Hypertension	Boolean	Yes	Yes/No
Chronic_Heart_Disease	Boolean	No	Yes/No
Chronic_Asthma	Boolean	No	Yes/No
Number_of_Chronic_Conditions	Numeric, Integer	2	
BMI	Numeric, Decimal	28.5	Body Mass Index
Smoking_Status	String	Former Smoker	Current, Former, Never
Allergies	String	Penicillin	Medical allergies
Risk_Score	Numeric, Decimal	7.5	Existing clinical risk score if available

Data request 4: Medication and treatment history

Column Name	Data Type	Example	Additional Information
-------------	-----------	---------	------------------------

Admission_ID	String	ADM987654	Links to admission
Medication_Name	String	Metformin	
Daily_Dosage	Numeric	500	Dosage amount
Number_of_Medications	Numeric, Integer	5	Total prescribed medications
Treatment_Completed	Boolean	Yes	
Follow_Up_Required	Boolean	Yes	Indicates mandatory follow-up

Data request 5: Follow-up and outpatient appointments

Column Name	Data Type	Example	Additional Information
Patient_ID	String	PAT123456	
Follow_Up_Date	Datetime	2025-05-15 09:00:00	Scheduled appointment
Attended	Boolean	No	
Days_to_Follow_Up	Numeric, Integer	8	Days between discharge and appointment
Missed_Appointments	Numeric, Integer	2	

Data request 6: Claim and billing information

Column Name	Data Type	Example	Additional Information
Claim_ID	String	CLM20250001	
Admission_ID	String	ADM123456	Links claim to admission
Patient_ID	String	PAT987654	
Hospital_ID	String	JHB001	
Claim_Submission_Date	Datetime	2025-05-15 14:30:00	Submission timestamp
Medical_Scheme	String	Discovery	

Plan_Type	String	Classic Saver	Medical scheme plan
Total_Claim_Value	Numeric, Decimal	45789.50	Total claim amount in ZAR
Patient_Copayment	Numeric, Decimal	1500.00	Out-of-pocket payment
Total_Line_Items	Integer	145	
Claim_Outcome	Categorical	Rejected	Approved, Queried, Short-Paid, Rejected
Settlement_Amount	Numeric, Decimal	39875.00	Amount paid by scheme
Query_Flag	Boolean	Yes	Indicates scheme query
Rejection_Reason	String	Incorrect coding	Populate if applicable

Data request 7: Procedure information

Column Name	Data Type	Example	Additional Information
Procedure_ID	String	PROC111	Unique procedure identifier
Admission_ID	String	ADM123456	
Procedure_Code	String	KNEE001	Standard procedure code
Procedure_Description	String	Total Knee Replacement	
Procedure_Date	Datetime	2025-05-02 09:30:00	
Surgeon_ID	String	DOC001	Anonymised surgeon identifier
Theatre_Time_Minutes	Integer	120	Length of procedure

Data request 8: Medical scheme rules (if available)

Column Name	Data Type	Example	Additional Information
Scheme_Name	String	Discovery	Medical scheme
Procedure_Code	String	KNEE001	
Preauthorisation_Required	Boolean	Yes	Requires approval
Maximum_Stay_Days	Integer	5	Approved hospital stay
Special_Rules	Text	Implant limit applies	Business rules